

EAG Global Buildathon • ShanHaiWoo Scholarship MVP PayAgent HSK

PayAgent HSK

Autonomous x402 Micropayments & Physical
Delivery-vs-Payment (DvP) on HashKey Chain

Protocol

x402 Machine Payments

Machine-to-machine HTTP 402 native payments. AI agents pay
APIs & services with zero human checkout friction.

Zero-Knowledge

Local Privacy Engine

Local selective disclosure credentials. Validates KYB and trade
secrets without broadcasting PII on public L2.

Physical AI

Robotic Fleet DvP

Escrow contracts unlocked by RISC-V hardware telemetry & GPS

coordinates for tokenized Colombian physical RWAs.

Pitch Script [0:00 - 0:20]:

"Good afternoon, judges. AI agents are here, but trapped in digital silos. We built PayAgent HSK: the financial infrastructure uniting Ethereum privacy, the x402 machine standard, and HashKey Chain L2 with physical real-world delivery."

HashKey Chain Testnet L2 (Chain ID: 133) Contract:
0x7A28cf37763279F774916b85b5ef8b64AB421f79 Slide 1 of 5

Problem Statement

The Trilemma of Autonomous Commerce

Why current AI agents are completely paralyzed in the enterprise & physical economy

1. The Payment Barrier

- Credit cards and bank accounts are hardcoded for human identities.
- Agents cannot pass 2FA SMS, CAPTCHAs, or 3D-Secure checkouts.
- Manual human-in-the-loop destroys 99% of agent automation.

Outcome: Trapped in simple chatbots.

2. Enterprise Privacy Fear

- Institutions refuse public L1/L2 because trading volumes leak on-chain.
- Corporate trade secrets & counterparty wallets are exposed to competitors.
- Strict regulatory compliance (KYB / Sanctions) lacks privacy rails.

Outcome: Zero enterprise blockchain adoption.

3. The Physical World Gap

- Smart contracts have no native awareness of physical delivery.
- Zero cryptographic trust between rovers, couriers, and buyers.
- High fraud & chargebacks in local physical commerce.

Outcome: Digital-only isolation.

Pitch Script [0:20 - 0:45]:

"Agents cannot hold credit cards. Enterprises fear leaking trade secrets on open block explorers. And no smart contract could verify if physical goods actually arrived on the street. We solved all three in one stack."

PayAgent HSK • Market Problem Analysis Slide 2 of 5

[Live Protocol Demo](#)

x402 Protocol & Local ZK Privacy

Machine-to-Machine HTTP payments meet
Zero-Knowledge selective disclosure

x402 Machine Payment Protocol (MPP)

How autonomous agents pay APIs in milliseconds without credit cards:

- **Step 1:** Agent requests restricted compute or inventory endpoint.
- **Step 2:** Server natively returns HTTP 402 Payment Required with L2-Pay price header in HSK.
- **Step 3:** Agent signs using delegated **ERC-4337 Session Key** on HashKey Chain.

- **Step 4:** Request re-sent with Payment-Proof header; access granted in < 2 seconds.

Local ZK Privacy Engine

Institutional confidentiality guaranteed:

- **Zero PII Broadcasted:** User credentials, ID cards, and bank balances remain in the local sandbox.
- **Selective Disclosure:** Generates a local cryptographic commitment keccak256(zkProof).
- **Verified On-Chain:** MerchantEscrow.sol validates proof without knowing trade secrets.

Pitch Script [0:45 - 1:45]:

"Look at the screen: our agent hits a locked resource, gets HTTP 402, and settles in milliseconds via its Session Key on HashKey Chain. Next, when buying tokenized Colombian coffee, the agent verifies local credentials with Zero-Knowledge proofs—zero private data leaked."

PayAgent HSK • Protocol Execution Slide 3 of 5

Physical Breakthrough

Robotic Fleet & Hardware DvP

The first physical Delivery-vs-Payment powered by autonomous agents on HashKey Chain

1. Escrow Lock

Customer orders Colombian Geisha Coffee (RWA). Funds locked securely in MerchantEscrow.sol on HashKey.

2. Rover Dispatch

Autonomous ground rover *CyberBot Medellín* navigates to delivery coordinates in Cali.

3. RISC-V Enclave

Onboard secure hardware enclave signs dual-GPS & sensor telemetry into an unforgeable hash.

4. DvP Release

Smart contract verifies telemetry hash and instantly releases HSK to the merchant wallet.

Pitch Script [1:45 - 2:30]:

"Here is the real magic: In our Physical AI Fleet, the rover navigates through Cali. Escrow funds NEVER release until the secure RISC-V hardware enclave signs the GPS and sensor telemetry. It is the world's first physical DvP on HashKey Chain!"

PayAgent HSK • Physical Delivery-vs-Payment Slide 4 of 5

Live Deployment & Impact

Production Ready on HashKey Chain L2

Live on-chain metrics, sub-cent fees, and full hackathon alignment

0.00007 HSK

Average Gas Fee (~\$0.0001 USD)

Enables high-frequency micro-payments unviable on Ethereum L1.

$\leq 2.0s$

Instant L2 Confirmation

Faster than Visa/Mastercard terminal handshakes for physical retail.

ERC-4337

Session Keys & Paymaster

Zero seed phrases. FaceID onboarding with sponsored gas abstraction.

Why We Win the Hackathon:

✓ **x402 Compliance:** 100% compliant with EAG workshop machine standards.

✓ **HashKey Ecosystem:** Utilizes HashKey Chain high throughput for institutional commerce.

✓ **Real Economic Impact:** Empowers local Colombian agricultural producers with tokenized RWAs.

Pitch Script [2:30 - 3:00]:

"PayAgent HSK is live right now on HashKey Testnet. We solve x402 standards, institutional privacy, and physical deliveries in one unified stack. The future of the agentic economy runs on HashKey Chain. Thank you!"

Presentation Deck: [Slide 1 / 5](#)

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